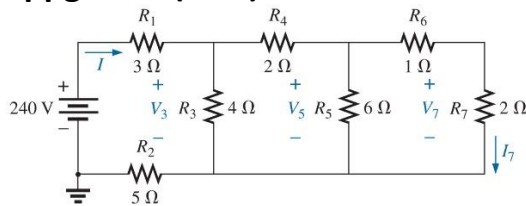


Løsningsforslag PB1120/PB2050

Vestfold – V22

Oppgave 1 (35%):



- Vis at $R_T = 10 \Omega$.
- Regn ut I (strømmen fra spenningskilden).
- Bruk strømdeling til å finne I_4 (strømmen gjennom R_4).
- Regn ut I_7 (strømmen gjennom R_7).
- Regn ut spenningene V_3 , V_5 og V_7 .
- Regn ut effekten levert av spenningskilden.
- Regn ut effekten levert til R_7 .

$$R'_T = R_5 \parallel (R_6 + R_7) = 6 \Omega \parallel 3 \Omega = 2 \Omega$$

$$R''_T = R_3 \parallel (R_4 + R'_T) = 4 \Omega \parallel (2 \Omega + 2 \Omega) = 2 \Omega$$

$$R_T = R_1 + R_2 + R''_T = 3 \Omega + 5 \Omega + 2 \Omega = 10 \Omega$$

$$I = \frac{240 \text{ V}}{10 \Omega} = \mathbf{24 \text{ A}}$$

$$I_4 = \frac{4 \Omega(I)}{4 \Omega + 4 \Omega} = \frac{4 \Omega(24 \text{ A})}{8 \Omega} = 12 \text{ A}$$

$$I_7 = \frac{6 \Omega(12 \text{ A})}{6 \Omega + 3 \Omega} = \frac{72 \text{ A}}{9} = \mathbf{8 \text{ A}}$$

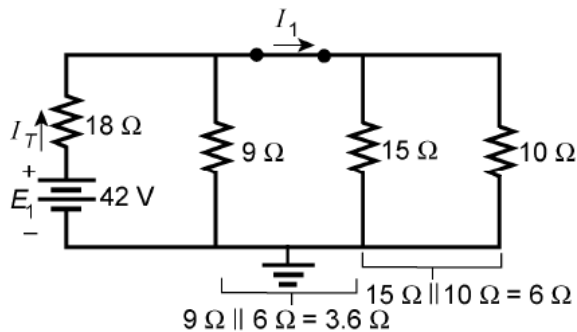
$$V_3 = I_3 R_3 = (I - I_4) R_3 = (24 \text{ A} - 12 \text{ A}) 4 \Omega = \mathbf{48 \text{ V}}$$

$$V_5 = I_5 R_5 = (I_4 - I_7) R_5 = (12 \text{ A} - 8 \text{ A}) 6 \Omega = \mathbf{24 \text{ V}}$$

$$V_7 = I_7 R_7 = (8 \text{ A}) 2 \Omega = \mathbf{16 \text{ V}}$$

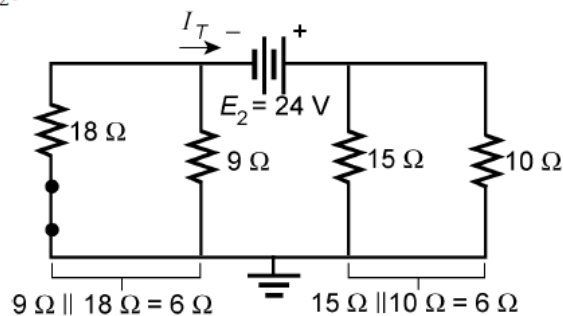
$$P = I_7^2 R_7 = (8 \text{ A})^2 2 \Omega = \mathbf{128 \text{ W}}$$

$$P = EI = (240 \text{ V})(24 \text{ A}) = \mathbf{5760 \text{ W}}$$

Oppgave 2 (15%): E_1 :

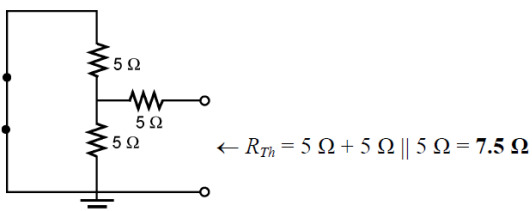
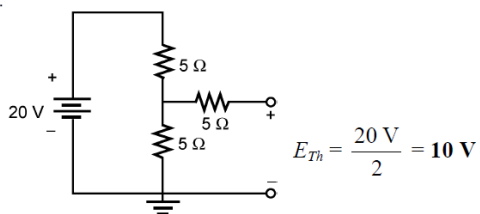
$$I_T = \frac{42\ \text{V}}{18\ \Omega + 3.6\ \Omega} = 1.944\ \text{A}$$

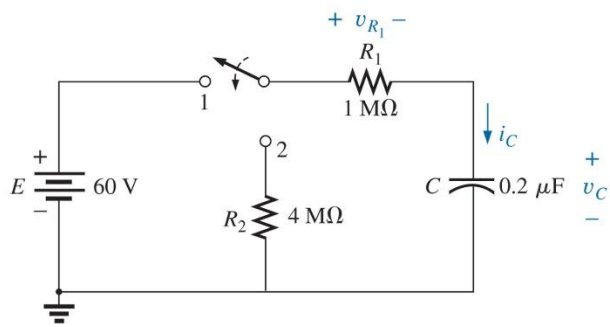
$$I_1 = \frac{9\ \Omega(I_T)}{9\ \Omega + 6\ \Omega} = \frac{9\ \Omega(1.944\ \text{A})}{15\ \Omega} = 1.17\ \text{A}$$

 E_2 :

$$I_T = \frac{E_2}{R_T} = \frac{24\ \text{V}}{12\ \Omega} = 2\ \text{A}$$

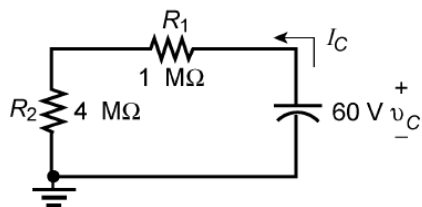
$$I_{24\text{V}} = I_T + I_1 = 2\ \text{A} + 1.17\ \text{A} = \mathbf{3.17\ \text{A (dir. of } I_1)}$$

Oppgave 3 (25%): R_{Th} : E_{Th} :

Oppgave 4 (25%):

Figur 4

- a) Etter 10 min i punkt 1 går det ingen strøm i kretsen ($i_C = 0$ A), og $v_C = E = 60$ V.
b)



$$\begin{aligned}\tau &= RC = (1 \text{ M}\Omega + 4 \text{ M}\Omega)(0.2 \text{ }\mu\text{F}) \\ &= (5 \text{ M}\Omega)(0.2 \text{ }\mu\text{F}) \\ &= 1 \text{ s}\end{aligned}$$

$$i_C = \frac{60 \text{ V}}{5 \text{ M}\Omega} e^{-t} = \mathbf{12 \text{ }\mu\text{A} e^{-t}}$$

$$8 \text{ }\mu\text{A} = 12 \text{ }\mu\text{A} e^{-t}$$

$$0.667 = e^{-t}$$

$$\log_e 0.667 = -t$$

$$-0.41 = -t$$

$$t = \mathbf{0.41 \text{ s}}$$